Alzheimer's Disease Progression Under Dropout: Longitudinal Modelling of BPRS and Severe CDR-SB

Moses Mburu

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Study Description

- Study design: Longitudinal Alzheimer cohort with repeated follow-up visits per patient (baseline + yearly visits).
- Outcomes:
  - Brief Psychiatric Rating Scale (BPRS) (continuous psychiatric symptom score)
  - Clinical Dementia Rating – Sum of Boxes (CDR-SB) severe indicator: $Z_{ij} = \mathbb{I}(\text{CDR-SB}_{ij} > 10)$
- Research questions
  - BPRS trajectory: Does BPRS change over time, and is its evolution associated with baseline characteristics (e.g., age/sex/education, baseline severity) and biomarkers (time-varying and/or baseline; e.g., ABPET/TAUPET)?
  - Severe cognition risk: Does the probability of severe cognition (CDR-SB > 10) increase over time, and is that increase associated with baseline variables and biomarkers?
- Challenge:
  - follow-up is incomplete due to dropout, with missingness increasing over time, so naïve complete-case analysis can be biased.

Methodology/Analysis of Data with missing data approaches

- Complete-case analysis (CC): analyse only subjects with complete follow-up (or only fully observed outcome vectors).
  - Simple, but can be biased unless missingness is MCAR (missing completely at random).
- last observation carried forward (LOCF): use all observed visits, and for dropout replace missing post-dropout values with the last observed value.
  - Uses more data than CC, but LOCF imposes a strong assumption that outcomes remain unchanged after dropout and can be biased if dropout is related to disease status.
- Multiple imputation (MI):
  - Impute missing outcomes multiple times from an imputation model and combine estimates.
  - Typically relies on MAR given the imputation model and included history/covariates.
- Likelihood-based models under MAR (primary approach):
  - Fit a longitudinal model (e.g., LMM/GLMM) by maximum likelihood (or REML for LMM).
- Weighted GEE (WGEE):
  - Extend GEE to handle dropout by weighting observed data by the inverse probability of being observed.
- Sensitivity analysis for MNAR:
  - Because MAR cannot be verified from the observed data alone, evaluate robustness under plausible departures using a pattern-mixture / δ-shift approach.

Results

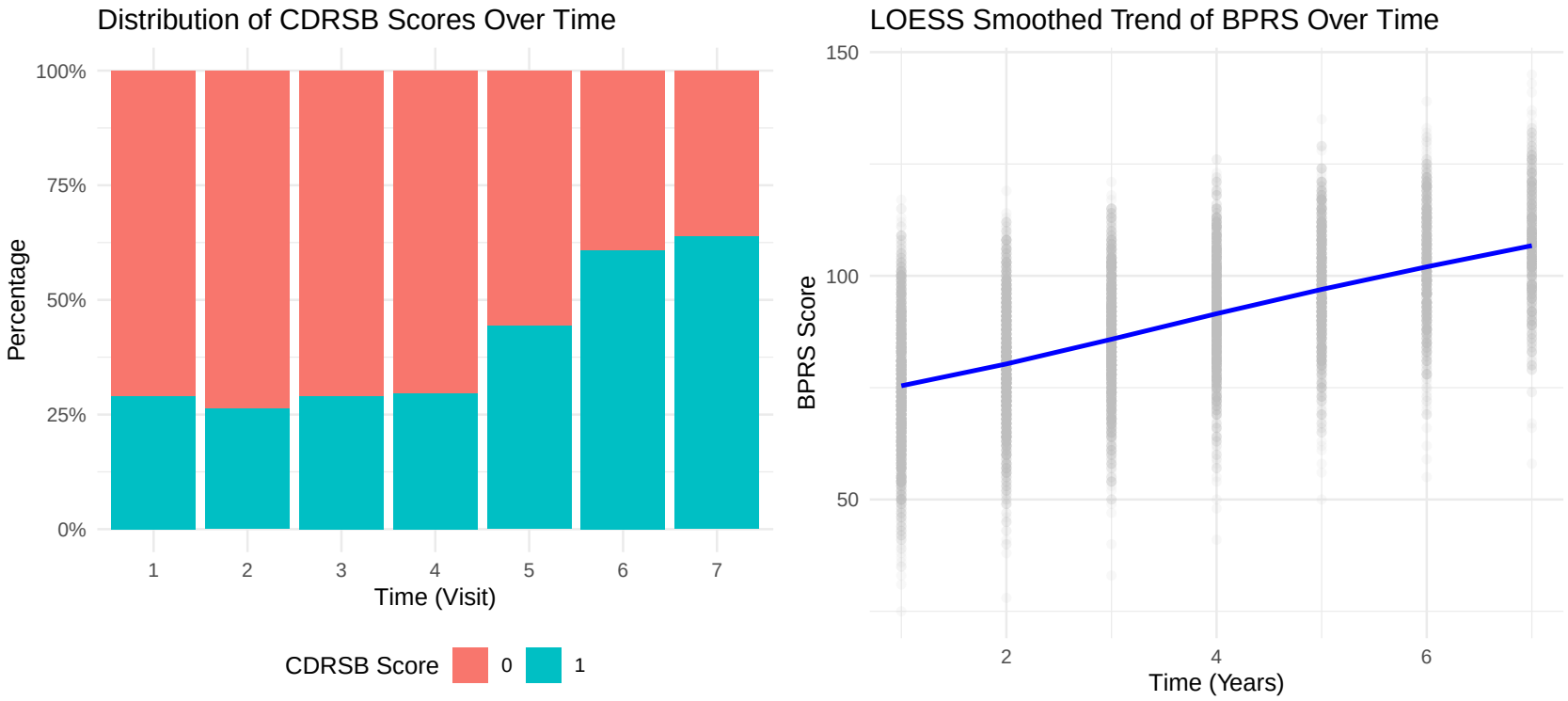
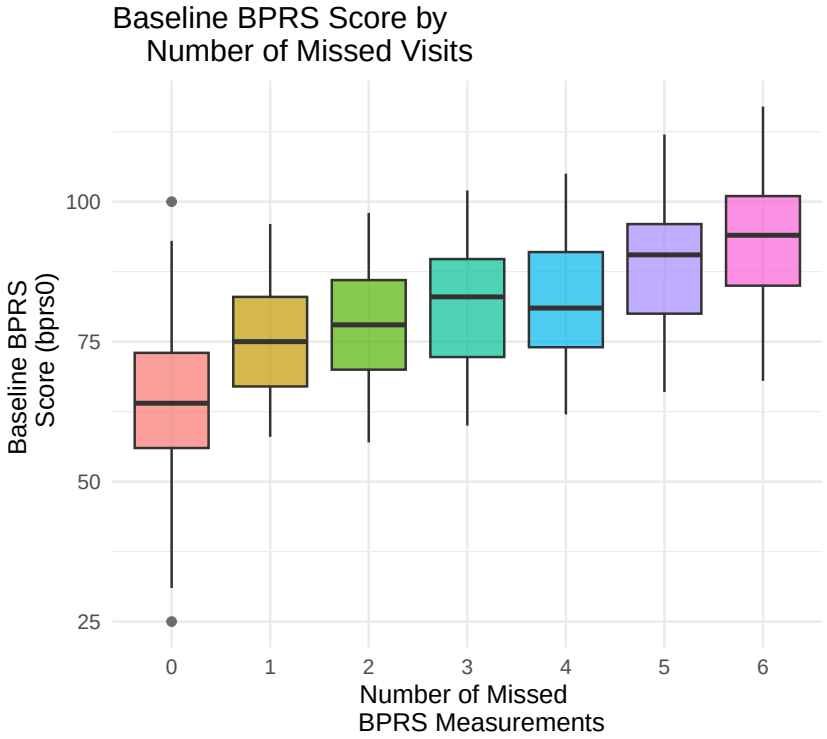
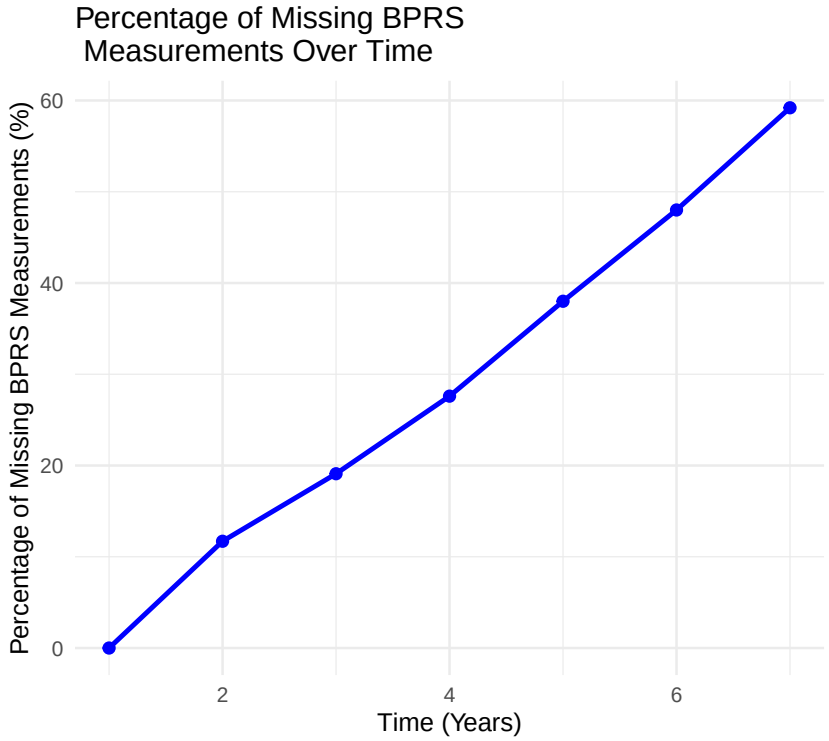


Figure 1: Distribution of CDRSB Scores & BPRS Over Time

The left panel shows that, as follow-up progresses, a growing proportion of patients move into the worse CDR-SB category (CDR-SB (>10)), indicating worsening cognition over time. The right panel shows a clear increasing LOESS trend in BPRS across years, suggesting psychiatric symptom burden also increases with time before we interpret these trends formally, we next examine missingness and dropout patterns.

Missing Data Patterns





- Missingness increases over time, reaching about 60% at the last follow-up.
- Participants with more missed visits have higher baseline BPRS, suggesting dropout is related to observed severity (not MCAR).

Table 1: Most common missingness patterns (O = observed, M = missing)

Missingness type	**Pattern (O=obs, M=miss)**	**n**	**%**
Complete follow-up	OOOOOOO	511	40.8
Monotone dropout	OOOOOOM	141	11.3
Monotone dropout	OOOOOMM	125	10.0
Monotone dropout	OOOOMMM	130	10.4
Monotone dropout	OOOMMMM	107	8.5
Monotone dropout	OOMMMMM	92	7.3
Monotone dropout	OMMMMMM	147	11.7

- The most common patterns show monotone dropout (once missing starts, later visits are usually missing).

Key findings (BPRS LMM + severe cognition models)

BPRS (LMM)

- BPRS increased over follow-up: +7.220 points/year (95% CI 7.141–7.299), $p < 0.001$
- Baseline factors associated with higher BPRS levels:
  - Older age ($p < 0.001$), higher BMI ($p < 0.001$), nursing home residence ($p < 0.001$)
  - Employment (paid job) was associated with lower BPRS ($p < 0.001$)
  - Sex was borderline (female vs male: $p = 0.067$)
  - ADL and baseline CDR-SB were not significant main effects
- Baseline severity moderated progression:
  - time $\times$ baseline CDR-SB $= -0.022$, $p < 0.001$
  - Interpretation: worse baseline CDR-SB $\rightarrow$ slower increase in BPRS over time (convergence/ceiling effect)

Severe cognition (CDR-SB > 10; GLMM)

- Risk increased with time: OR $= 1.986$ per year (95% CI 1.783–2.211), $p < 0.001$
- Biomarker modification:
  - time $\times$ tau PET OR $= 1.473$, $p = 0.020$ higher tau accelerates progression to severe status
- Other covariates were not significant in this subject-specific model:
  - sex, age, BMI, ADL, amyloid PET, and tau PET main effect (all $p > 0.05$)

Conclusion (what the clinician needs to hear)

- Models estimate how BPRS and severe cognition (CDR-SB > 10) change over time and which patient factors are associated with faster worsening, helping identify higher-risk patients for closer follow-up.
- Sensitivty Analysisδ-shift? When we assumed post-dropout ($\delta = 0$ to 3) shift, the main conclusions did not change: the time trend remained strongly positive and the key predictors (age, BMI, nursing home, paid job, and the time $\times$ baseline CDR-SB effect) stayed consistent, while weaker effects (sex, ADL, baseline CDR-SB main effect) were not robust.